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Test Pattern



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Main)

AIOT

13-02-2021

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	D	B	A	C	C	C	D	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	D	A	C	C	A	D	A	A
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	0.63	11.70	2.00	0.38 to 0.42	2.50	5.00	18.00	9.10	0.10	0.09

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	C	C	A	B	D	C	D	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	C	C	A	A	D	D	D	A	D
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	4.00	68.00 TO 68.50	1.00	8.00	0.80	3.00	5.00	122.00	8.40 TO 8.57	8.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	C	D	D	C	D	A	A	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	D	D	A	B	A	C	C	D
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	0.00	11.00	6.00	0.36	0.75	0.25	0.25	0.58 or 0.59	0.25	0.75



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PART-1 : PHYSICS

SOLUTION

SECTION-I

1. Ans. (B)

Sol. From Gauss's Law

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

$$E \cdot 4\pi r^2 = \frac{\int_0^r (4\pi r'^2 \cdot dr') (\alpha r)}{\epsilon_0}$$

$$E \cdot 4\pi r^2 = \frac{4\pi \cdot \alpha \cdot r^4}{4 \epsilon_0}$$

$$E = \left(\frac{\alpha r^2}{4 \epsilon_0} \right)$$

2. Ans. (B)

Sol. Since plane is smooth hence motion is only translational

$$a = g \sin \theta = 10 \times \frac{1}{2} = 5 \text{ m/s}^2$$

$$s = ut + \frac{1}{2} at^2$$

$$1.6 = 0 + \frac{1}{2} \times 5 \times t^2$$

$$t^2 = \sqrt{\frac{3.2}{5}} = \sqrt{0.64} = 0.8 \text{ sec.}$$

3. Ans. (D)

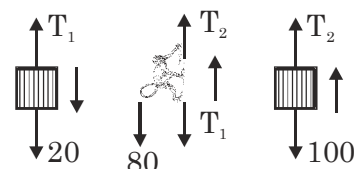
$$\text{Sol. } \frac{P}{Q} = \frac{R}{X} \quad X_{\max} = \frac{R_{\max} \cdot Q_{\max}}{P_{\min}}$$

$$= 9000 \times \frac{1000}{10} = 9 \times 10^5 \text{ ohm}$$

$$X_{\min} = \frac{R_{\min} \cdot Q_{\min}}{P_{\max}} = 100 \times \frac{10}{1000} = 1 \text{ ohm}$$

4. Ans. (B)

Sol. Let the acceleration of string = a



$$\text{For 2 kg } 20 - T_1 = 2a \dots\dots\dots(i)$$

$$\text{For monkey } T_2 - 80 - T_1 = 8(2 - a) \dots\dots(ii)$$

$$\text{For 10 kg } T_2 - 100 = 10a \dots\dots\dots(iii)$$

$$\therefore a = 0.8 \text{ m/s}^2$$

$$\text{Acceleration of monkey w.r. to ground} \\ = 2 - 0.8 = 1.2 \text{ m/s}^2$$

$$s = ut + \frac{1}{2} at^2$$

$$2.4 = 0 + \frac{1}{2} \times 1.2 \times t^2$$

$$t = 2 \text{ sec.}$$

5. Ans. (A)

Sol. If $n_2 \rightarrow n_1$ in H(Z = 1) gives λ then $zn_2 \rightarrow zn_1$ give same λ H-likeion for He^+ ion (z = 2)

6. Ans. (C)

Sol. From conservation of Angular momentum about hinged point

$$mv\ell = \left(\frac{m\ell^2}{3} + 2m \cdot \ell^2 \right) \omega$$

$$\omega = \left(\frac{3}{7} \frac{v}{\ell} \right)$$

Now from conservation of Energy.

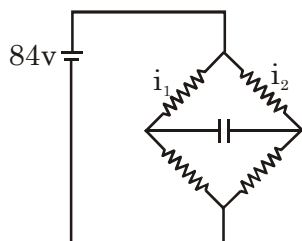
$$\frac{1}{2} \times \frac{7}{3} m\ell^2 \times \omega^2 = mg\ell + 2mg \cdot 2\ell$$

$$v = \sqrt{\frac{70}{3} g\ell}$$

7. Ans. (C)

Sol. $i_1 = \frac{84}{140} = 0.6A$

$i_2 = \frac{84}{60} = 1.4A$

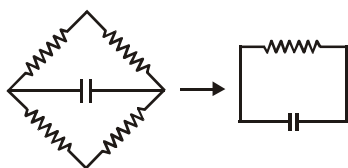


p.d across capacitor = 20 v

$Q = CV = 5 \times 20 = 100 \mu C$

After S is open

$Q = Q \cdot e^{-t/RC}$
 $= 100 e^{-4}$



8. Ans. (C)

Sol. $\frac{1}{2} \mu v_r^2 = \frac{1}{2} kx^2$

$x = V_{rer}^2 \sqrt{\frac{\mu}{k}}$

$= 2 \times \sqrt{\frac{8/6}{10}} = \sqrt{\frac{8}{15}}$

From conservation of momentum

$2 \times 4 + 4 \times 2 = 2v_1 + 4v_2$

$8 = v_1 + 2v_2 \quad \dots(1)$

From conservation of energy

$\frac{1}{2} \times 2 \times 4^2 + \frac{1}{2} \times 4 \times 2^2 = \frac{1}{2} \times 2 \times v_1^2 + \frac{1}{2} \times 4 \times v_2^2$

... (ii)

On solving $v_1 = \frac{4}{3} \text{ m/s}, v_2 = \frac{10}{3} \text{ m/s}$

9. Ans. (D)

Sol. Area of ellipse = πab

Magnetic dipole moment = $\frac{\pi abi}{2}$

Torque $\vec{\tau} = \vec{M} \times \vec{B}$

$\tau = MB \sin \theta$

$\therefore \vec{M} \parallel \vec{B} \quad \boxed{\tau = 0}$

10. Ans. (C)

Sol. To reach at point A particle must cross the peak point.

Loss in KE = gain in PE

$\frac{1}{2} \times \frac{100}{1000} \times v^2 = (5 - 0)$

$v^2 = 100$

$v = 10 \text{ m/s}$

11. Ans. (A)

Sol. Here the width of principal maxima is 2.5 mm

Then its half width is = $\frac{\beta}{2}$

$= \frac{2.5}{2} = 1.25 \times 10^{-3} \text{ m}$

Diffraction angle $\theta = \frac{\left(\frac{\beta}{2}\right)}{D} = \frac{1.25 \times 10^{-3}}{2} \quad \dots(i)$

$\therefore a\theta = \lambda \quad \therefore \theta = \frac{\lambda}{a} \quad \dots(ii)$

From (i) and (ii)

$\lambda = \theta a$

$= \frac{1.25 \times 10^{-3}}{2} \times 10^{-3}$

$= 6.25 \times 10^{-7} \text{ m}$

$\therefore \lambda = 6250 \text{ \AA}$

12. Ans. (B)

Sol. $\frac{1}{0.9} = + \frac{1}{(10-2t)} + \frac{1}{2t}$

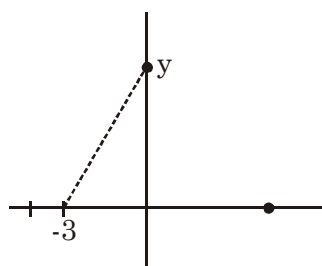
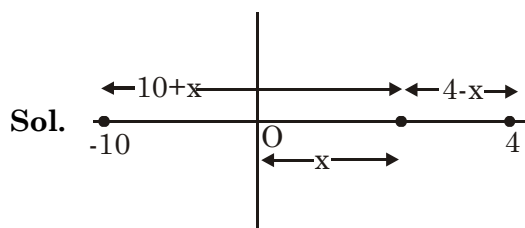
$t = 0.5 \text{ sec and } t = 4.5 \text{ sec.}$

13. Ans. (D)

Sol. Stopping potential increases and number of photons decreases.

$$I = \frac{hc n_p}{\lambda t}$$

14. Ans. (A)



$$t = \frac{10+x}{C} = \frac{4-x}{C}$$

$$\therefore -6 = 2x$$

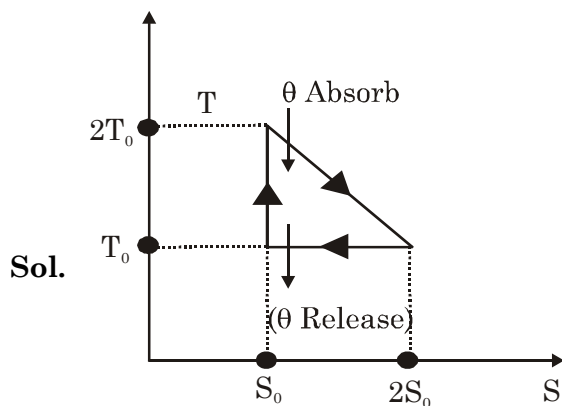
$$t = \frac{10-3}{C} \quad \therefore x = -3$$

$$\therefore t = \frac{7}{C}$$

$$\therefore 7 = \sqrt{3^2 + y^2}$$

$$\therefore y = \sqrt{40}$$

15. Ans. (C)



$$\eta = \frac{\text{W.D. by gas}}{\text{Heat absorb}}$$

$$\eta = \frac{\text{Heat(Net)}}{\text{Heat(Absorb)}}$$

$$\eta = \frac{\text{Area under the curve}}{\text{Heat Absorb}}$$

$$\eta = \frac{\frac{1}{2} \times S_0 T_0}{\frac{1}{2} (3T_0) S_0} = \frac{1}{3}$$

16. Ans. (C)

Sol. The time period of LC oscillations,

$$T = 2\pi\sqrt{LC}$$

The time at which charge on the capacitor

will be zero is $\frac{T}{4}$.

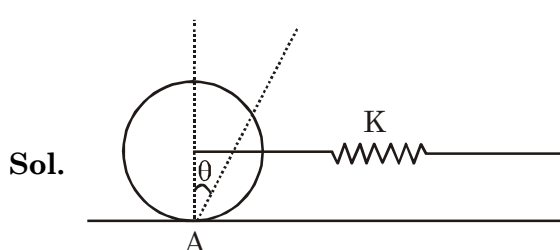
$$\text{So } t = \frac{\pi}{2}\sqrt{LC}$$

17. Ans. (A)

Sol.
$$I = \frac{E}{R_2}$$

$$U_{\text{Inductor}} = \text{Heat loss in } R_1 = \frac{1}{2} L \left(\frac{E}{R_2} \right)^2$$

18. Ans. (D)



$$\tau_A = I\alpha$$

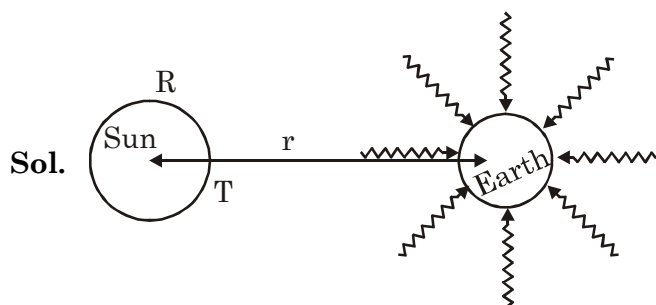
$$K \times R = \frac{3}{2} MR^2 \alpha$$

$$KR\theta R = \frac{3}{2} MR^2 \alpha$$

$$\alpha = \frac{2K}{3M}$$

$$\therefore T = 2\pi\sqrt{\frac{3M}{2K}}$$

19. Ans. (A)



Sol.

$$\frac{e A_{\text{sun}} \sigma T^4}{4 \pi r^2} \times A_{\text{proj Earth}}$$

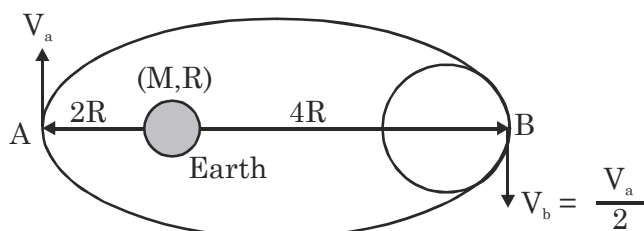
$$\frac{4 \pi R^2}{4 \pi r^2} \sigma T^4 \pi r_0^2$$

$$\pi \frac{R^2}{r^2} r_0^2 \sigma T^4$$

20. Ans. (A)

Sol. $\frac{1}{2} m V_a^2 - \frac{GMm}{2x}$

$$= \frac{1}{2} m \frac{V_a^2}{4} - \frac{GMm}{4x}, \quad V_a = \sqrt{\frac{2}{3} \frac{GM}{x}}$$

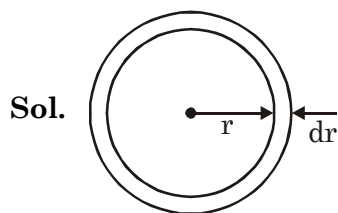


ROC at point - B

$$ROC = \frac{V_b^2}{a \perp} = \frac{\frac{2GM}{3R} \times \frac{1}{4}}{\frac{GM}{(4R)^2}} = \frac{8R}{3}$$

SECTION-II

1. Ans. 0.63



$$\tau = \int F \cdot r = \int_0^R \eta (2 \pi r dr) \left(\frac{rw}{t} \right) \cdot r$$

$$= \frac{2 \pi \eta W R^4}{4 t} = \frac{2 \times 3.14 \times 1 \times 8 \times 10^{-4}}{4 \times 2 \times 10^{-3}} = 0.625 \text{ N-m} \approx 0.63 \text{ N-m}$$

2. Ans. 11.70

Sol. $I = \frac{12 - 0.3}{5 \times 10^3} = 2.34 \text{ mA}$

$$V_0 = IR = (2.34 \times 11^{-3}) (5 \times 10^3) = 11.7 \text{ V}$$

3. Ans. 2.00

Sol. $n(7800) = (n+1) 5200$

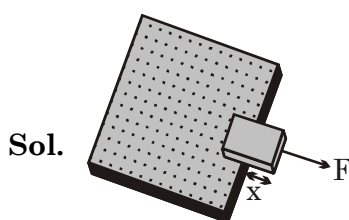
4. Ans. 0.38 to 0.42

Sol. $\Delta L = L_0 \alpha \Delta T$
 $= 12 \times 11 \times 10^{-6} \times 30$
 $= 3960 \times 10^{-6} \text{ m}$

5. Ans. 2.5 J

Sol. $\frac{\pi^2 a^2 F}{4L} = \pi^2 \times \frac{1}{\pi^2} \frac{10}{4 \times 1} = 2.5 \text{ J}$

6. Ans. 5.00



In this case,

$$F = \eta \frac{v}{h} \ell (\ell - x)$$

$$\therefore v = \frac{Fh}{\eta \ell (\ell - x)} = \frac{dx}{dt}$$

$$\text{So, } t = \frac{3 \eta \ell^3}{8 F h} = 5 \text{ sec}$$

7. Ans. 18.00

Sol. $5 \times 10^3 I_1 = 60$

$$\Rightarrow I_1 = \frac{60}{5 \times 10^3} = 12 \text{ mA}$$

Potential difference across

$$10 \text{ k}\Omega = 240 - 60 = 180 \text{ V}$$

$$\text{So, } I_3 \times 10 \times 10^3 = 180$$

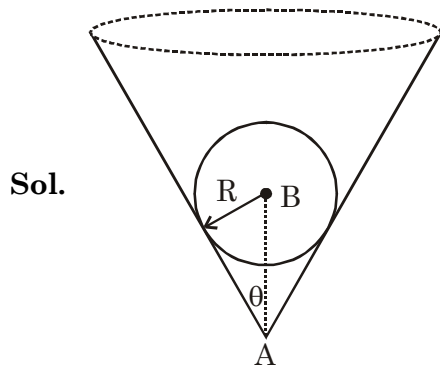
$$\Rightarrow I_3 = 18 \text{ mA}$$

$$I_2 = I_3 - I_1 = 6 \text{ mA}$$

8. Ans. 9.10

Sol. $7 \times \frac{D \lambda}{d} = \frac{(\mu_1 - \mu_2) t \cdot D}{d}$
 $t = 9.1 \text{ }\mu\text{m}$

9. Ans. 0.10



Sol.

$$\theta = 30^\circ$$

$$\therefore AB = \frac{R}{\sin \theta} = 2R$$

Also, height of cone $H = 3R \cot \theta$

$$\therefore \Delta U = mg\Delta(2R) + mg\Delta\left(\frac{2H}{3}\right)$$

But $\Delta R = R\alpha\Delta T$ and $\Delta H = H\alpha\Delta T$

Putting values,

$$\Delta U = +0.1 \text{ J}$$

10. Ans. 0.09

$$\text{Sol. Energy in 1 loop} = \int \frac{1}{2} dm [A \sin(kx)\omega]^2$$

$$= \frac{\lambda A^2 \omega^2}{8} \left(\frac{M}{L} \right)$$

$$\therefore \text{Ratio} = \frac{3}{5} \times \frac{3}{5} \times \frac{1}{4}$$

PART-2 : CHEMISTRY

SOLUTION

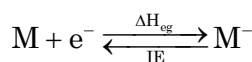
SECTION-I

1. Ans. (D)

Sol. Neighboring group participation by a sulphide involved.

2. Ans. (B)

Sol. $\Delta H_{eg}(M) = IE(M^-)$



Since sulphur has most -ve electron gain enthalpy hence S^- will have highest IE.

3. Ans. (C)

$$\text{Sol. } r = r_0 \frac{n^2}{Z}$$

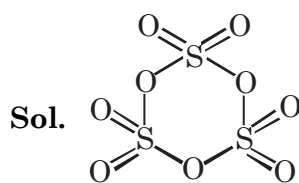
As Z increases, r_0 decreases

Hence upon increasing Z , nuclear distance of node decreases.

4. Ans. (C)

Sol. Process of active methylene group $\uparrow$ acidic strength and the side chain should be e^- withdrawing.

5. Ans. (A)



Sol.

S_3O_9 has no
S-S linkage

6. Ans. (B)

Sol. $A \rightarrow 2B + C$

$$t = 0 \quad a \quad 0 \quad 0$$

$$t = 10 \quad a - x \quad 2x \quad x$$

$$t \rightarrow \infty \quad 0 \quad 2a \quad a$$

$$a + 2x \propto 176$$

$$3a \propto 270$$

$$a \propto 90$$

$$\therefore 2x \propto 86$$

$$x \propto 43$$

$$a - x \propto 90 - 43$$

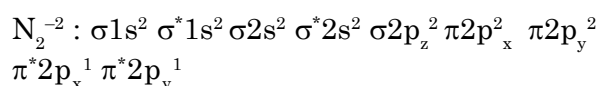
$$a - x \propto 47$$

7. Ans. (D)

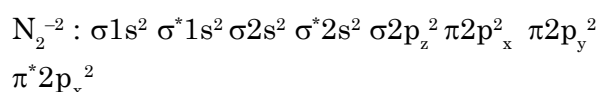
Sol. Yield of 4-hydroxypentan-2-one is low because aldol and diacetone alcohol are also formed.

8. Ans. (C)

Sol. If Hund's rule is followed



If Hund's rule is not followed

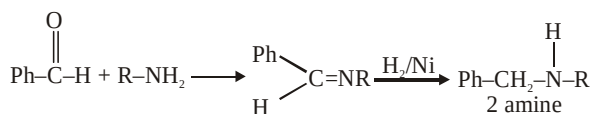


9. Ans. (D)

Sol. Fe acts as catalyst and Mo acts as promoter in Haber's process.

10. Ans. (C)

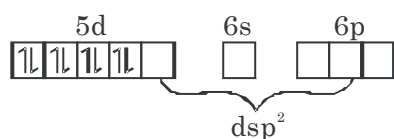
Sol.



11. Ans. (C)

Sol. (A) $[\text{IrCl}(\text{CO})(\text{PPh}_3)_2]$ carbonylchloridobis (triphenylphosphine) iridium(I).

(B) Coordination number of Ir is four. Ir is in (+1) oxidation state with $4d^8$ configuration. It is trans isomer, so its geometry should be square planar.



(C) All electrons are paired, hence diamagnetic

(D) Can exhibit GI only.

12. Ans. (C)

Sol. Anode $\text{Ag} + \text{Br}^- \rightarrow \text{AgBr} + \text{e}^-$

Cathode $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$

$$E^\circ_{\text{cell}} = (E^\circ_{\text{RP/C}}) - (E^\circ_{\text{RP/A}})$$

$$= E^\circ_{\text{Ag}^+|\text{Ag}} - \left[E^\circ_{\text{Ag}^+|\text{Ag}} + 0.06 \log K_{\text{SP}} \right]$$

$$E^\circ_{\text{cell}} = -0.06 \log K_{\text{sp}}$$

$$Q = \frac{1}{[\text{Br}^-][\text{Ag}^+]} = \frac{1}{0.1 \times 0.1} = 100$$

$$E_{\text{cell}} = -0.06 \log K_{\text{sp}} - 0.06 \log 100$$

$$0.643 = -0.06 \log K_{\text{sp}} - 0.12$$

$$\log K_{\text{sp}} = -\frac{0.763}{0.06} = -12.71$$

13. Ans. (C)

Sol. Fernald Craft reaction not shown by Highly Activated Rings.

14. Ans. (A)

Sol. The slag formed is CaSiO_3 which is lighter and has less melting point.

15. Ans. (A)

Sol. $\text{AgBr} \rightleftharpoons \text{Ag}^+ + \text{Br}^-$
 $\text{S} + 10^{-7} \quad \text{S}$

$$(\text{S} + 10^{-7})\text{S} = 12 \times 10^{-14}$$

$$\therefore \text{S} = 3 \times 10^{-7} \text{ M}$$

$$[\text{Ag}^+] = 4 \times 10^{-7} \text{ M}$$

$$[\text{Br}^-] = 3 \times 10^{-7} \text{ M}$$

$$[\text{NO}_3^-] = 10^{-7} \text{ M}$$

$$\kappa_{\text{total}} = [6 \times 4 \times 10^{-7}] + [8 \times 3 \times 10^{-7}] + [7 \times 1 \times 10^{-7}]$$

$$= 55 \times 10^{-7} \text{ S m}^{-1}$$

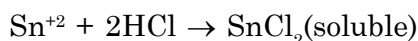
16. Ans. (D)

Sol. DiBAL-H is electron deficient reducing agent as Burane and readily reduces electron rich sp.

17. Ans. (D)

Sol. A, B and C can be precipitated by HCl as well as H_2S as their insoluble chlorides and sulphides.

(D) Sn^{+2} can be precipitated by H_2S but not by HCl



18. Ans. (D)

Sol. Kiliani Fischer method is for upgradation of C-chain.

19. Ans. (A)

Sol. At Boyle's Temperature; gas behaves ideally for a range of pressure.

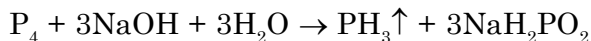
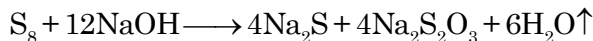
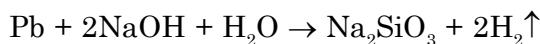
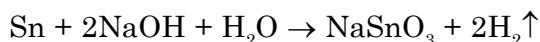
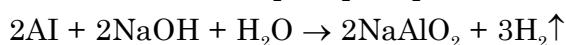
20. Ans. (D)

Sol. All are correct

SECTION-II

1. Ans. 4.00

Sol. $\text{Zn} + 2\text{NaOH} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2\uparrow$



2. Ans. 68.00 to 68.50

Sol. Let V ml is taken

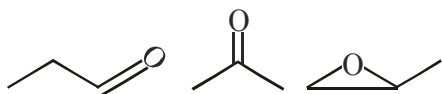
$$\frac{5 \times V + 1 \times 150}{(V + 150) \times 0.9} = 2.5$$

$$5V + 150 = 2.25V + 337.5$$

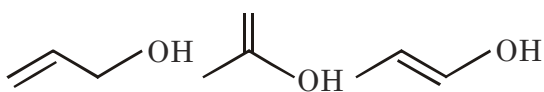
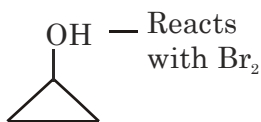
$$V = \frac{187.5}{2.75} = 68.18 \text{ ml}$$

3. Ans. (1.00)

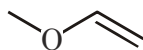
Sol. Molecular formula from calculation comes to be C_3H_6O i.e., it stands for the following compounds.



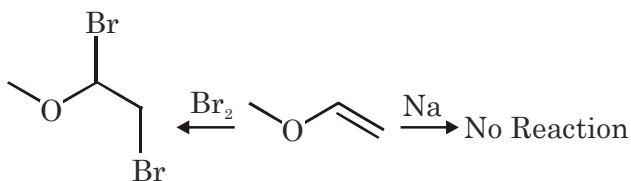
No addition reaction with Br_2



Reacts with Na



$\therefore$ is the compound



$$VD \times 2 = MM \quad = 29 \times 2 = 58$$

$\therefore$ 58 g of ether combines with $2 \times 80g Br_2$

$\therefore$ 2.9 g of ether will combine with

$$\frac{2 \times 80}{58} \times 2.9 = 8g \text{ of Bromine}$$

$$Y = 9 \text{ and } X = 8$$

$$Y - X = 9 - 8 = 1$$

4. Ans. (8.00)

- Sol. (i) give NO_2 (ii) NH_4NO_3
 (iii) gives NO_2 (iv) Give NO_2
 (v) give NO_2
 (vi) Given N_2O (vii) Gives NO_2
 (viii) Gives NO_2 (ix) Gives NO

5. Ans. (0.80)

Sol. $C_V = \frac{3R}{2}$

AC polytropic $P \propto V$

$$\Rightarrow P.V^{-1} = \text{constant}$$

$$C_{AC} = C_V + \frac{R}{1 - (-1)} = 2R$$

AB isobaric

$$C_{AB} = C_P = \frac{5R}{2}$$

$$\therefore \frac{Q_{AC}}{Q_{AB}} = \frac{C_{AC}}{C_{AB}} \quad [\because \Delta T \text{ same for both}]$$

$$= \frac{2R}{\left(\frac{5R}{2}\right)} = \frac{4}{5} = 0.8$$

6. Ans. (3.00)

Sol. $x = 4$

$$y = 6$$

$$z = 6$$

Ans. (3)

7. Ans. (5.00)

Sol. N_2O , C_2H_2 , CO_2 , C_3O_2 , BeF_2

8. Ans. (122.00)

9. Ans. (8.40 to 8.57)

10. Ans. (8.00)

SECTION-I

1. Ans. (C)

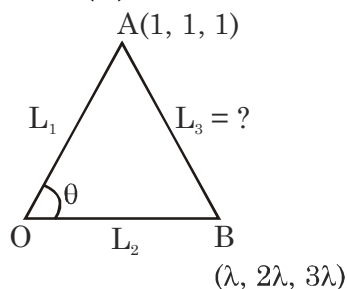
Sol. First element in 20th group is equal to '20'.
(20, 21, upto 39 times)

$$\therefore \text{Sum} = \frac{39}{2}(2 \times 20 + (39-1) \times 1)$$

$$= \frac{39}{2}(40 + 38) = (39)^2$$

2. Ans. (B)

Sol.



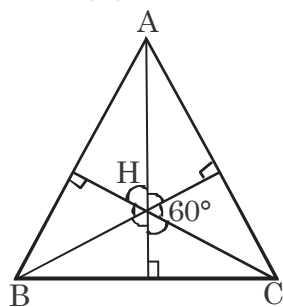
$$\cos \theta = \frac{6}{\sqrt{42}} \Rightarrow \sin \theta = \frac{\sqrt{6}}{\sqrt{42}} \quad (\text{where } \theta \text{ is the angle between } L_1 \text{ \& } L_2)$$

$$\text{Area } \triangle OAB = \frac{1}{2}(OA)(OB)\sin \theta$$

$$= \frac{1}{2}(\sqrt{3})|\lambda|(\sqrt{14})\frac{\sqrt{6}}{\sqrt{42}} = \sqrt{6} \Rightarrow \lambda = \pm 2$$

3. Ans. (C)

Sol.



The two altitudes are

$$(x - \sqrt{3}y + 1)(x + \sqrt{3}y - 5) = 0$$

Point of int. of the 2 altitudes is $(2, \sqrt{3})$

Let slope of 3rd altitude be 'm'

$$\text{then } \left| \frac{m - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}} \right| = \sqrt{3} \Rightarrow \frac{\sqrt{3}m - 1}{\sqrt{3} + m} = \pm \sqrt{3}$$

$$\sqrt{3}m - 1 = \pm 3 \pm \sqrt{3}m$$

$$m = \infty, m = \frac{-2}{2\sqrt{3}} = \frac{-1}{\sqrt{3}}$$

$\therefore$ The third altitude is

$$x = 2$$

4. Ans. (D)

$$\text{Sol. } 56 = \frac{50n + 60m}{m + n}$$

$$56n + 56m = 50n + 60m$$

$$6n = 4m$$

$$\frac{n}{m} = 2/3$$

$$\therefore 9 \cdot \frac{n}{m} = 9 \times \frac{2}{3} = 6$$

5. Ans. (D)

Sol. Centre of C_3 will lie on the radical axis of C_1 and C_2 which is $10x + 6y + 26 = 0$.

Let center of C_3 is (h, k) .

Equation of chord of contact through (h, k)

to the C_1 may be given as $hx + ky = 25$... (I)

Let the mid point of the chord is (x_1, y_1)

the equation of the chord with the help of mid point may be given as

$$xx_1 + yy_1 = x_1^2 + y_1^2$$

Since (I) and (II) represents same straight line $10x + 6y + 26 = 0$

$$\Rightarrow \frac{h}{25} = \frac{x_1}{x_1^2 + y_1^2}, \frac{k}{25} = \frac{y_1}{x_1^2 + y_1^2}$$

Since (h, k) lie on the radical axis

$$\Rightarrow 10\left(\frac{25x}{x_1^2 + y_1^2}\right) + 6\left(\frac{25y_1}{x_1^2 + y_1^2}\right) + 26 = 0$$

The locus of (x_1, y_1) is

$$5x + 3y + \frac{13}{25}(x^2 + y^2) = 0$$

6. Ans. (C)

Sol. The equation of the line through $P(x, y)$ making an angle with the x-axis which is supplementary to the angle made by the tangent at $P(x, y)$ is

$$Y - y = -\frac{dy}{dx}(X - x) \quad \dots (1)$$

At the point where it meets the x-axis

$$Y = 0, X = x + \frac{y}{\frac{dy}{dx}} \Rightarrow OA = x + \frac{y}{\frac{dy}{dx}} \quad \dots (2)$$

The line through P(x,y) and perpendicular to (1) is

$$Y - y = \frac{dx}{dy}(X - x).$$

At the point where it meets the y-axis

$$X = 0, Y = y - \frac{x}{\frac{dy}{dx}} \Rightarrow OB = y - \frac{x}{\frac{dy}{dx}}.$$

Since OA = OB,

$$x + \frac{y}{\frac{dy}{dx}} = y - \frac{x}{\frac{dy}{dx}} \Rightarrow (y - x) = \frac{y + x}{\frac{dy}{dx}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y + x}{y - x}$$

$$y dy - x dy = y dx + x dx$$

$$y dy - x dx = x dy + y dx$$

$$\int y dy - \int x dx = \int d(xy)$$

$$\frac{y^2}{2} - \frac{x^2}{2} = xy + c$$

$$\frac{1}{2} - \frac{1}{2} = 1 + c$$

$$c = -1$$

$$\therefore y^2 - x^2 = 2xy - 2$$

$$\boxed{x^2 + 2xy - y^2 = 2}$$

7. **Ans. (D)**

$$\text{Sol. } \sim(\sim q \wedge p) = (\sim(\sim q)) \vee (\sim p)$$

$$= \sim p \vee q$$

$$= p \rightarrow q$$

$\therefore$ contrapositive is $\sim q \rightarrow \sim p$

Hence, if $\sim q$ then $\sim p$

8. **Ans. (A)**

$$\text{Sol. } A^2 |A|^2 - |B|^{3/2} B$$

$$= 25A^2 - |B| \cdot B$$

$$= 25A^2 + |B|A^2$$

$$= 0$$

9. **Ans. (A)**

$$\text{Sol. In } x \in (-2, 2)$$

$||x| - 1|$ is not differentiable at $x = -1, 0, 1$

$|\cos \pi x|$ is not differentiable

$$\text{at } x = -\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}$$

10. **Ans. (A)**

$$\text{Sol. } a_0 C_0 - a_1 C_1 + a_2 C_2 - a_3 C_3 + \dots + a_{2012} C_{2012}$$

$$= a_0 (C_0 - C_1 + C_2 + \dots + C_{2012})$$

$$- d (C_1 - 2C_2 + 3C_3 - 4C_4 + \dots - {}^{2012}C_{2012})$$

$$= a_0 (0) - d (C_1 - 2C_2 + 3C_3 - 4C_4$$

$$+ \dots - {}^{2012}C_{2012})$$

$$= a_0(0) - d(0)$$

$$= 0$$

11. **Ans. (B)**

$$\text{Sol. } |x| + |y| \leq \frac{1}{2}$$

$$\text{If } x = y, \quad 2|x| \leq \frac{1}{2}, \quad -\frac{1}{4} \leq x \leq \frac{1}{4}$$

$$\therefore (x, x) \notin R \quad \forall x \in \text{real no.}$$

$\therefore R$ is not reflexive

$$\text{If } |x| + |y| \leq \frac{1}{2} \Rightarrow |y| + |x| \leq \frac{1}{2}$$

$$\therefore (x, y) \in R \Rightarrow (y, x) \in R$$

$\therefore R$ is symmetric

$$\text{If } |x| + |y| \leq \frac{1}{2} \text{ and } |y| + |z| \leq \frac{1}{2}$$

$$\not\Rightarrow |x| + |z| \leq \frac{1}{2}$$

$\therefore R$ is not transitive.

12. **Ans. (B)**

$$\text{Sol. } |\vec{a} + \vec{b} + \vec{c} + \vec{d}| = \sum |\vec{a}|^2 + 2 \sum \vec{a} \cdot \vec{b}$$

$$= 4 + 2\vec{d} \cdot (\vec{a} + \vec{b} + \vec{c}) \quad (\text{a, b, c are mutually } \perp \text{ r})$$

$$\text{Let } \vec{d} = \lambda \vec{a} + \mu \vec{b} + \nu \vec{c}$$

$$\text{Then } \vec{d} \cdot \vec{a} = \vec{d} \cdot \vec{b} = \vec{d} \cdot \vec{c} = \cos \theta$$

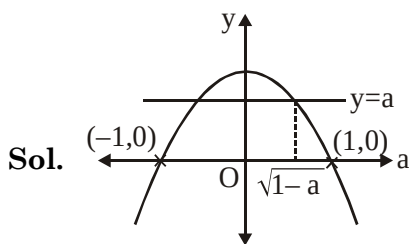
$$\therefore \lambda = \mu = \nu = \cos \theta$$

$$\text{Also } \lambda^2 + \mu^2 + \nu^2 = 1 = 3 \cos^2 \theta \text{ or } \cos \theta = \frac{1}{\sqrt{3}}$$

$$\therefore |\vec{a} + \vec{b} + \vec{c} + \vec{d}|^2 = 4 \pm 2 \cdot \frac{3}{\sqrt{3}}$$

$$= 4 \pm 2 \frac{3}{\sqrt{3}}$$

13. Ans. (D)



$$A(a) = 2 \int_0^{\sqrt{1-a}} ((1-x^2) - a) dx = \frac{4}{3} (1-a)^{3/2}$$

$$\therefore A(0) = \frac{4}{3}$$

$$\text{and } A\left(\frac{1}{2}\right) = \frac{4}{3} \left(\frac{1}{2}\right)^{3/2} \Rightarrow \frac{A(0)}{A\left(\frac{1}{2}\right)} = 2\sqrt{2}$$

14. Ans. (D)

Sol. $(a-1)x + 0y + z = \alpha \quad \dots(1)$

$$x + (b-1)y + 0z = \beta \quad \dots(2)$$

$$0x + y + (c-1)z = \gamma \quad \dots(3)$$

For no unique solution $D = 0$

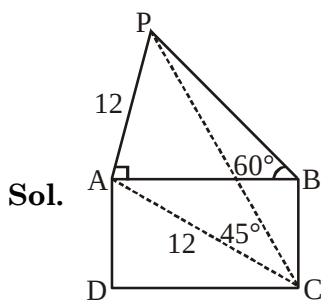
$$\begin{vmatrix} a-1 & 0 & 1 \\ 1 & b-1 & 0 \\ 0 & 1 & c-1 \end{vmatrix} = 0$$

$$(a-1)(b-1)(c-1) + 1 = 0$$

$$\therefore a=2; b=2; c=0$$

$$\text{Hence, } |a+b+c| = 4$$

15. Ans. (A)



$$\triangle APC, AC = AP = 12$$

$$\triangle ABP, \tan 60^\circ = \frac{12}{AB}$$

$$AB = \frac{12}{\sqrt{3}} = 4\sqrt{3}$$

$$BC^2 = AC^2 - AB^2 = 144 - 48 = 96$$

$$BC = \sqrt{96} = 4\sqrt{6}$$

$$\text{Area} = 4\sqrt{3} \cdot 4\sqrt{6} = 48\sqrt{2}$$

16. Ans. (B)

Sol. $\cos^3 3x + \cos^3 5x = (2 \cos 4x \cos x)^3$

$$= (\cos 5x + \cos 3x)^3$$

$$\cos^3 3x + \cos^3 5x$$

$$= \cos^3 5x + \cos^3 3x + 3 \cos 5x \cos 3x (\cos 5x + \cos 3x)$$

$$\Rightarrow (3 \cos 3x \cdot \cos 5x) (2 \cos 4x \cdot \cos x) = 0$$

$$\Rightarrow \cos x \cdot \cos 3x \cdot \cos 4x \cdot \cos 5x = 0$$

$$\therefore x = (2n+1)\frac{\pi}{2}, (2n+1)\frac{\pi}{6}, (2n+1)\frac{\pi}{8}, (2n+1)\frac{\pi}{10}$$

$$\Rightarrow \text{Smallest +ve values of } x \text{ is } \frac{\pi}{10} \text{ i.e. } 18^\circ.$$

17. Ans. (A)

Sol. Note that $D^r < 0$, hence given inequality

$$(1) \text{ is true only if } N^r \geq 0$$

$$\text{i.e. } (x-8)(x-2) \leq 0 \text{ and } 2^{x-3} > 31$$

$$\text{i.e. } 2 \leq x \leq 8 \text{ and } 2^{x-3} \geq 31$$

$$\text{only } x = 8 \text{ satisfies both the inequality.}$$

18. Ans. (C)

Sol. $P(W) = \frac{1}{3}; P(B) = \frac{2}{3}$

$$\Rightarrow p = \frac{1}{3}; q = \frac{2}{3} \text{ and } r = 4 \text{ or } 5 \text{ and } n = 5$$

$$\text{Use } P(r) = {}^nC_r p^r q^{n-r}$$

$$P(4) + P(5)$$

$$= {}^5C_4 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^1 + {}^5C_5 \left(\frac{1}{3}\right)^5$$

$$= \frac{10}{3^5} + \frac{1}{3^5} = \frac{11}{3^5} = \frac{11}{243}$$

19. Ans. (C)

Sol. $\therefore \arg(z) = \pi \text{ if } z = x + iy$

$$\therefore y = 0 \text{ and } x < 0$$

$$\therefore |z + 3 - i| = 1 \Rightarrow |x + 3 - i| = 1$$

$$\Rightarrow (x+3)^2 + 1 = 1 \Rightarrow x = -3$$

$$\therefore z = -3 + i \cdot 0 \Rightarrow |z| = 3$$

20. Ans. (D)

Sol. In the integral J, substitute $x + 1 = t$

$$\Rightarrow dx = dt \text{ and } x^2 + 2x = (t^2 - 1)$$

$$\text{Now } J = \int_1^e \frac{e^{\frac{t^2-2}{2}}}{t} dt \text{ and } K = \int_1^e t \ln t e^{\frac{t^2-2}{2}} dt$$

$$\text{Hence } (J + K) = \int_1^e e^{\frac{t^2-2}{2}} \left(\frac{1}{t} + t \ln t \right) dt$$

$$= \left(e^{\frac{t^2-2}{2}} \ln t \right)_{t=1}^{t=e} = e^{\frac{e^2-2}{2}} = (\sqrt{e})^{e^2-2}$$

SECTION-II

1. Ans. (0.00)

Sol. Let P is a root ($\in I$)

Case-I : p is odd

$$p^{10} + ap^9 + b \neq 0$$

Because LHS odd

Case-II : p is even

$$p^{10} + ap^9 + b \neq 0$$

because, LHs is odd

2. Ans. (11.00)

Sol. Out of numbers $\frac{1}{2}, \frac{2}{4}, \frac{3}{6}, \frac{1}{3}, \frac{2}{6}, \frac{2}{3}, \frac{4}{6}$ will result only 3 distinct rational numbers.

$$\Rightarrow \text{Total numbers} = {}^6C_2 - 7 + 3 = 11$$

3. Ans. (6.00)

Sol. Any point on $x + y = 1$ can be taken as $(t, 1-t)$

The equation of chord with this as mid-point is $y(1-t) - 2a(x+t) = (1-t)^2 - 4at$

It passes through $(a, 2a)$

$$\text{So, } t^2 - 2t + 2a^2 - 2a + 1 = 0$$

This should have two distinct real roots.

So $D > 0$

$$\Rightarrow a^2 - a < 0$$

$$\Rightarrow 0 < a < 1$$

So, length of latus rectum < 4 and $0 < a < 1$

$$\Rightarrow \text{LR} = 1, 2, 3$$

4. Ans. (0.36)

$$\text{Sol. } \lim_{x \rightarrow 0} \left[\sin^2 \left(\frac{\pi}{2-3x} \right) \right]^{\sec^2 \left(\frac{\pi}{2-5x} \right)}$$

$$e^{\lim_{x \rightarrow 0} \left[\sin^2 \left(\frac{\pi}{2-3x} \right) - 1 \right] \sec^2 \left(\frac{\pi}{2-5x} \right)}$$

$$e^{\lim_{x \rightarrow 0} \frac{-\cos^2 \left(\frac{\pi}{2-3x} \right)}{\cos^2 \left(\frac{\pi}{2-5x} \right)}} = e^{\lim_{x \rightarrow 0} \frac{-\sin^2 \left(\frac{\pi}{2} - \frac{\pi}{2-3x} \right)}{\sin^2 \left(\frac{\pi}{2} - \frac{\pi}{2-5x} \right)}}$$

$$e^{\lim_{x \rightarrow 0} \frac{-\sin^2 \left(\frac{-3\pi x}{2(2-3x)} \right)}{\sin^2 \left(\frac{-5\pi x}{2(2-5x)} \right)}} = e^{\frac{9}{25}}$$

5. Ans. (0.75)

$$\text{Sol. } |\vec{a} \times (\vec{a} \times \vec{c})| = |3\vec{b}| = 3|\vec{b}|$$

$$|\vec{a}| \cdot |(\vec{a} \times \vec{c})| \sin \frac{\pi}{2} = 3.1 \Rightarrow 3 = |\vec{a}| \cdot (|\vec{a}| |\vec{c}| \sin \theta)$$

$$\Rightarrow 3 = 3.2 \sin \theta \Rightarrow \sin \theta = \frac{1}{2} \Rightarrow \cos^2 \theta = \frac{3}{4}$$

6. Ans. 0.25

$$\text{Sol. } L = \lim_{x \rightarrow \infty} \frac{\left(2 + \frac{1}{x} \right)^{40} \left(4 - \frac{1}{x} \right)^5}{\left(2 + \frac{3}{x} \right)^{45}} = 32$$

7. Ans. 0.25

$$\text{Sol. } \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) \Rightarrow a + b = 2$$

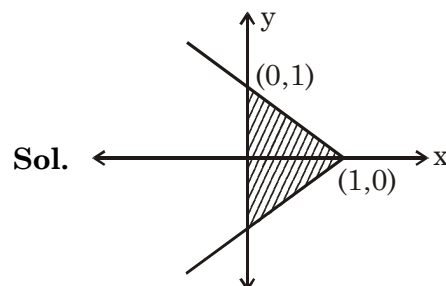
$$\therefore \lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x) \Rightarrow 3a + b = 6 \tan \frac{3\pi}{12}$$

$$\therefore a = 2, b = 0$$

8. Ans. 0.58 or 0.59

$$\text{Sol. } \int_0^{1.5} [x^2] dx = \int_0^1 0 dx + \int_1^{\sqrt{2}} 1 dx + \int_{\sqrt{2}}^{1.5} 2 dx = 2 - \sqrt{2}$$

9. Ans. 0.25



Sol.

$$\text{area} = 2 \times \left(\frac{1}{2} \times 1 \times 1 \right) = 1 = k$$

10. Ans. 0.75

$$\text{Sol. } k = \frac{[\vec{a} \vec{b} \vec{c}] + 2[\vec{a} \vec{b} \vec{c}] + [\vec{a} \vec{b} \vec{c}] - [\vec{a} \vec{b} \vec{c}]}{[\vec{a} \vec{b} \vec{c}]}$$

$$k = 3$$